



AI reinvention made real

Accenture Innovation Challenge 2026

Team details

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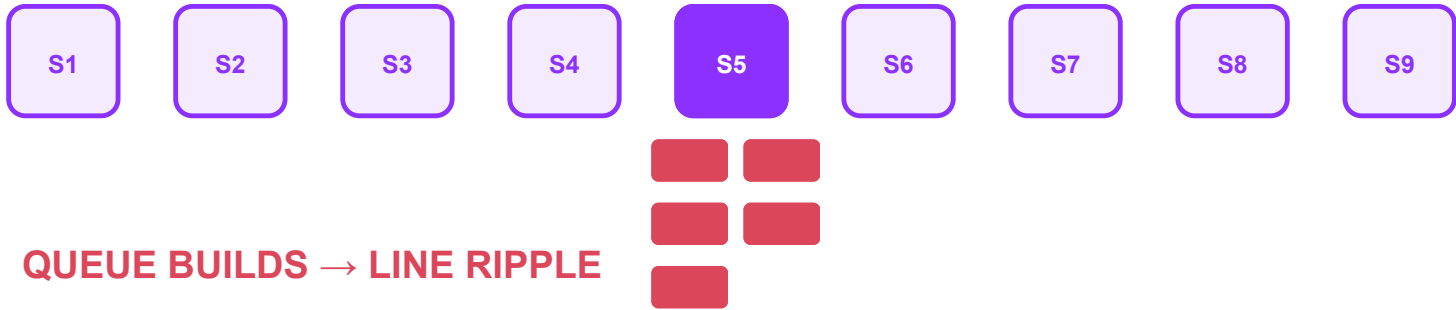


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Problem Statement

VEHICLE ASSEMBLY LINE

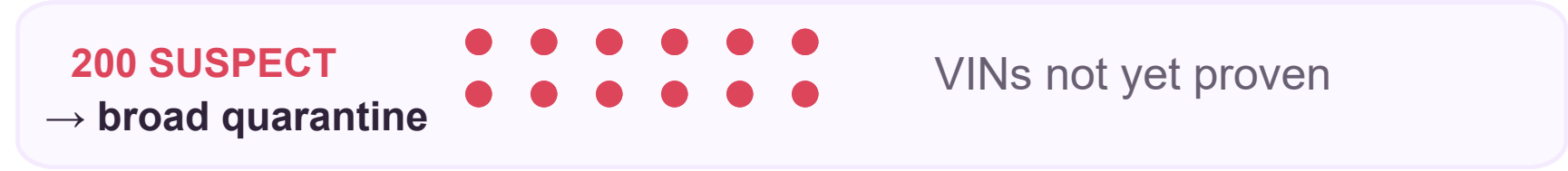


A vehicle assembly line is a chain of interdependent stations — welding, painting, engine fit, and final inspection. When one station slows even slightly, units queue upstream while stations downstream idle waiting for input, degrading the rhythm of the entire line.

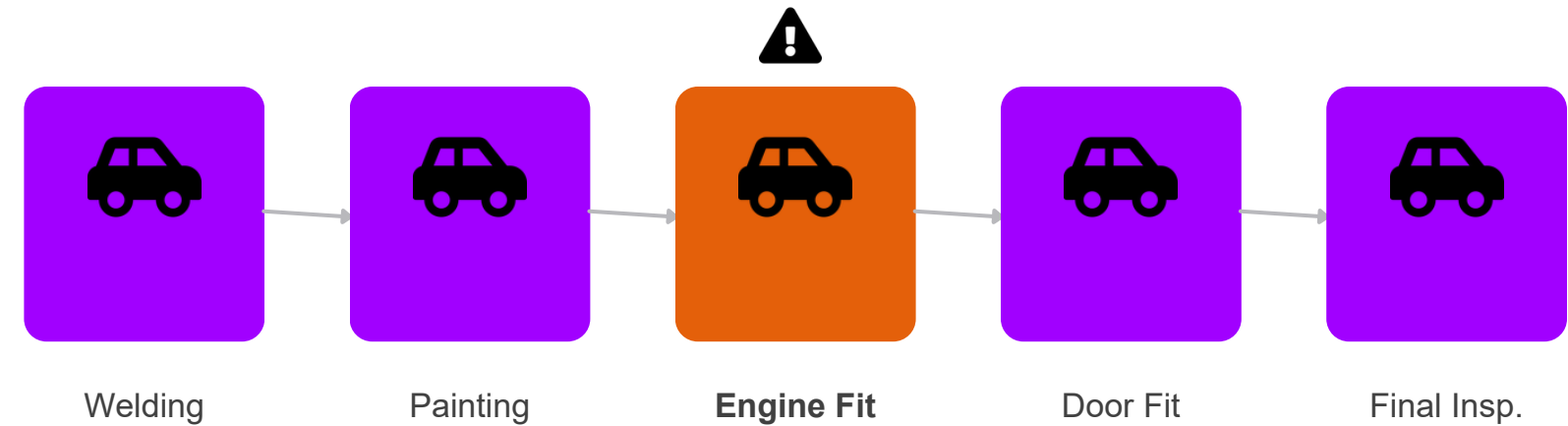
Early-stage defects: a mis-torqued bolt, a drifting weld parameter - are rarely caught at their source. By the time final inspection flags them, dozens of vehicles carrying the same flaw have already moved through, turning a small deviation into costly rework or a recall.

DEFECTS SPREAD BEFORE THEY'RE KNOWN

A tool drifts toward its limit → vehicles keep passing → alarm arrives too late.



Where one slow station breaks the whole line



BOTTLENECKS ARE FOUND TOO LATE

One station starts slipping → queues grow → downstream stations starve.

WHAT HAPPENS



~35% OF STATIONS UNMONITORED

manual stations • legacy machines • no retrofit

The true constraint may be in the part of the line nobody can see.

THE CORE GAP

Reactive monitoring → Late detection → Lost throughput + broad quarantine

Digital Twin: Predict defects before they happen



VIN / Barcode

Vehicle identity + timestamps



PLC / Machine

States + cycle events



Process sensors

Torque • temp • pressure



Quality records

Inspection + outcomes



Station events

Entry • exit • waits



DIGITAL TWIN ENGINE

MODEL sequence, cycle/wait, queues, key parameters, VIN history

SKIP full 3D physics, machine-level detail, new hardware

MVP | EARLY PROCESS PREDICTION

- 1. Drift detection** 50 → 49.5 → 49 Nm; predicts a limit breach about 2 hrs early.
- 2. Change-point check** confirms a real shift before alerting, fewer false alarms.

COMPLETE | PER-VIN DEFECT PREDICTION

- 3. Per-VIN history** tracks VIN 4471 across stations 8→15→22, not station silos.
- 4. Cross-station risk** low torque @8 + thick sealant @15 → high rattle risk.

VALID EVEN WITHOUT SENSOR DATA

No sensor at a station? The twin estimates the missing signal from VIN timestamps, PLC states, and queue position, reported as an estimate with a confidence range, never treated as a direct measurement. Predictions start conservative and sharpen automatically as labeled outcomes accumulate.



ONE prioritized alert, not three: “Vehicle 4471 - High defect risk. Inspect before final assembly.”

“Station 8: torque breach predicted 3:05 PM. Recalibrate at 1:00 break.”



Prevent the defect before it happens



Zero unplanned downtime



If one escapes: 200 → 12 quarantined, with proof

Catch the Bottleneck Before the Line Stops

DOWNTIME EXPOSURE \$2.3M / hour

DigitalTwin.ai | Blind-station observability → predictive bottleneck detection → prescriptive control

SEE THE BLIND SPOTS

**~35% of stations
may be unmonitored**

Existing signals are unified: no
PLC retrofit required.

VIN / gate timestamps

Smart tools telemetry

Edge vision* station pose

ANDON FILTER

Pause timers during global line
stops

PREDICT THE CONSTRAINT

DIGITAL TWIN

station activity + buffer dynamics

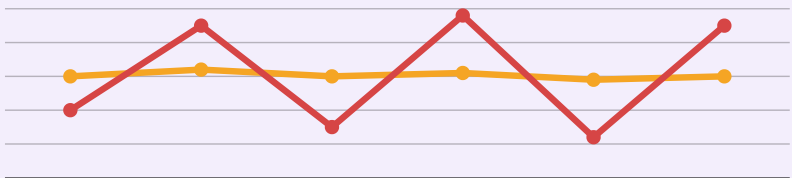
VARIANCE BOTTLENECK

Buffer swing → true constraint

PACING-PRESSURE RISK

>85% takt load → defect risk rises

BUFFER SIGNAL



ACT BEFORE THE STALL

- HOLD** Option-heavy VIN clusters
upstream
- FORECAST** Stall + recovery window
- ESCALATE** 5 min → supervisor | 45 min
→ plant
- VERIFY** Targeted EOL check for
high-risk VINs

PREVENT • DON'T JUST ALERT

WHY IT WINS

NO RETROFIT

FEWER FALSE ALARMS

PREDICTIVE CONTROL

Video | DARKLINE — 3-minute solution walkthrough

CORTEX · IIT Kanpur · Problem Statement 4: DigitalTwin.ai

Video link: [\[Click Here for link to video on Google Drive\]](#)

or copy from here: https://drive.google.com/drive/folders/1KURY9-vGq_73xce399jnJM7l8YIVectD?usp=sharing

ATTACHED VIDEO:



**Kindly use the google drive link attached above in case the video doesn't play in the slides.*

Thank you

